CADzilla

Board User Reference Manual

S-100 bus graphics controller board — Hitachi HD63484 ACRTC & Brooktree Bt453 video DAC

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1. Overview

CADzilla is an S-100 bus board built around the Hitachi HD63484 Advanced CRT Controller (ACRTC) paired with a Brooktree Bt453 video DAC. The board carries 2 MBytes of video frame buffer memory and supports 8-bit-per-pixel (256-color) display modes at resolutions up to 1024×768.

The board occupies a single, jumper/switch-selectable 8-byte block of S-100 I/O space. This manual documents that I/O map, the board's MODE register, the SW1 configuration DIP switch, and the set of HD63484 features the board does and does not support.

2. I/O Address Map

The board decodes a single 8-byte-aligned base I/O address, set by SW1 (see Section 4). All board registers are addressed as offsets from that base address:

Offset	Register	Access
`Base + 0`	HD63484 Address/Status Register (RS = 0)	Read/Write
`Base + 1`	MODE Register (board configuration — see Section 3)	Write only
`Base + 2`	HD63484 Control/FIFO Register (RS = 1)	Read/Write
`Base + 3`	Reserved / not decoded	—
Base + 4 – Base + 7	Bt453 video DAC registers	Read/Write

RS is the HD63484 register-select input: RS = 0 selects the address/status register at Base + 0, RS = 1 selects the control/FIFO register at Base + 2.

3. MODE Register (Base + 1, write only)

The MODE register configures board-level options that are outside the HD63484's own register set. It is write-only.

Bit	Name	Description
0	HSPOL	Horizontal sync polarity: 0 = positive sync pulse, 1 = negative sync pulse
1	VSPOL	Vertical sync polarity: 0 = positive sync pulse, 1 = negative sync pulse
2	AMODE	Address mode: 0 = single access mode, 1 = interleaved access mode
3	OLEN	Overlay enable (TBD) — set to 0 for now
4–7	—	Reserved, write as 0

4. Configuration DIP Switch (SW1)

SW1 is an 8-position DIP switch that sets the board's I/O base address and host interface options.

Switch	Function	On	Off
SW1-1 – SW1-5	I/O base address (address bits 7:3)	—	—
SW1-6	16-bit host mode select	HD63484 set to 16-bit host interface mode at reset	HD63484 set to 8-bit host interface mode at reset
SW1-7	16-bit address decode enable	Decodes full 16-bit I/O address	Decodes only 8-bit I/O address
SW1-8	Interrupt enable	HD63484 interrupt request connected to the S-100 bus	Interrupt request disconnected from the S-100 bus

SW1-1 through SW1-5 map directly to address bits 7 down to 3 of the board's I/O base address; the low three address bits (2:0) are fixed at zero, so the board's registers always occupy an 8-byte-aligned block (Base + 0 through Base + 7) somewhere in the 256-byte S-100 8-bit I/O space.

5. Frame Buffer

Property	Value
Memory type	2 MBytes of SRAM — no refresh cycles required
Pixel depth	8 bits per pixel only
Graphics Address Increment (GAI)	8
Maximum display resolution	1024 × 768

6. HD63484 Feature Support Summary

CADzilla implements a deliberately reduced subset of the HD63484's capabilities, matched to the board's fixed 8-bit-per-pixel, SRAM-based frame buffer. The table below summarizes what is and is not supported.

Feature	Status
Pixel depth	8 bits per pixel only (GAI = 8)
Frame buffer refresh	Not required (SRAM frame buffer)
Cursors	Not supported
Superimposed (dual-screen) access mode	Not supported
Light pen	Not supported
Zooming	Not supported
Video attributes	Not supported
External sync	Not supported
DMA transfers	Not supported
Interrupts	Supported (enabled via SW1-8, see Section 4)

7. Revision History

Revision	Date	Notes
1.0	2026-09-23	Initial I/O map, MODE register, SW1, and feature support documentation.